

Erdős Problem #39

EP-39

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Is there an infinite Sidon set $A \subset \mathbb{N}$ such that $|A \cap \{1 \dots, N\}| \gg_{\epsilon} N^{1/2-\epsilon}$ for all $\epsilon > 0$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>